

MACKAY MO, AustraliaWMO: **943670**Lat: **21.1172S**Lon: **149.2169E**Elev: **119**StdP: **14.63**Time Zone: **10.00 (E10)**Period: **95-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	49.7	52.1	35.3	30.3	59.0	39.2	35.4	58.9	22.6	66.1	20.8	65.7	9.0	230	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	89.8	78.1	88.0	77.4	86.6	76.6	80.6	86.2	79.5	84.8	78.6	83.7	12.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	151.2	83.7	78.0	146.2	82.8	77.1	141.9	82.0	44.4	86.5	43.2	85.1	42.3	83.8	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.9	19.0	44.2	93.0	3.1	2.0	42.0	94.4	40.2	95.6	38.5	96.7	36.2	98.2		
(5)			40.5	82.6	2.8	1.8	38.5	83.9	36.9	84.9	35.4	85.9	33.4	87.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.1	80.5	80.0	78.3	74.5	69.3	64.9	63.4	65.1	69.9	74.6	77.5	80.0
	DBStd	6.94	2.50	2.30	2.46	2.65	3.59	4.14	4.04	3.80	3.59	3.24	2.76	2.82
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	191	0	0	0	0	10	52	78	46	5	0	0	0
	CDD50	8442	946	841	878	735	598	448	415	467	597	762	825	930
	CDD65	3156	481	421	413	285	142	50	28	47	152	297	375	465
	CDH74	24025	4831	4126	3515	1500	322	46	27	115	581	1636	2831	4495
	CDH80	5342	1296	1029	639	111	11	2	1	5	62	278	626	1282
(14) Wind	WSAvg	10.5	11.1	11.2	11.7	12.0	10.9	10.5	9.6	9.2	9.2	10.1	10.2	10.5
(15) Precipitation	PrecAvg	62.6	10.8	12.4	11.3	5.6	4.2	2.5	1.6	1.0	0.8	1.4	3.6	7.1
	PrecMax	112.1	40.7	42.2	36.3	20.5	21.5	15.4	10.6	4.8	9.2	6.0	22.2	45.6
	PrecMin	31.3	0.6	1.1	1.4	0.4	0.6	0.1	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	20.7	8.9	9.2	8.9	3.9	3.2	2.5	1.7	1.1	1.3	1.5	4.3	7.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.5	91.3	88.8	85.3	81.2	78.3	77.0	80.5	84.4	87.2	89.1	92.3
		MCWB	79.4	80.7	79.3	76.2	72.2	70.3	68.4	69.6	71.7	74.7	75.3	78.0
	2%	DB	89.1	88.0	86.4	82.7	78.9	75.0	73.8	77.1	81.8	85.0	87.0	89.7
		MCWB	78.5	79.0	77.8	74.3	70.9	68.4	66.7	66.7	70.3	73.2	74.9	77.4
	5%	DB	87.3	86.4	84.7	81.0	77.1	73.2	71.9	74.5	79.5	83.1	85.4	87.9
		MCWB	77.7	77.7	76.4	73.4	69.9	67.6	65.8	65.7	69.3	72.3	74.0	76.7
	10%	DB	85.7	85.2	83.4	79.7	75.6	71.7	70.3	72.4	77.3	81.1	83.7	86.0
		MCWB	76.8	77.1	75.6	72.8	69.0	66.6	64.7	65.3	68.1	71.2	73.1	75.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.9	82.5	80.9	78.2	75.2	72.8	70.9	71.6	74.1	76.9	78.3	81.0
		MCDB	87.8	88.5	86.3	82.4	78.5	75.6	74.1	77.1	80.5	84.7	85.7	88.0
	2%	WB	80.4	80.5	79.3	76.4	73.0	70.7	69.1	69.5	72.5	75.2	76.8	79.3
		MCDB	86.1	85.7	84.0	80.0	76.8	73.2	71.8	73.7	78.2	81.3	83.7	85.9
	5%	WB	79.2	79.2	78.2	75.2	71.6	69.3	67.8	68.3	71.3	74.1	75.6	78.2
		MCDB	84.5	83.9	82.6	79.1	75.6	72.0	70.2	72.4	76.6	80.3	82.4	84.5
	10%	WB	78.3	78.4	77.2	74.0	70.3	68.0	66.5	67.0	70.1	72.8	74.6	77.2
		MCDB	83.4	82.9	81.4	78.2	74.3	70.8	69.4	71.0	75.1	79.0	81.2	83.4
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	9.7	9.8	10.3	12.4	12.8	14.3	15.4	15.1	13.3	12.2	11.8
		MCDBR	12.9	12.0	11.8	12.0	13.7	13.5	15.4	18.3	18.2	15.8	14.6	14.7
		MCWBR	5.8	5.7	5.8	6.3	7.8	8.5	9.7	10.5	8.8	6.8	5.9	5.8
	5% WB	MCDBR	11.4	11.0	10.6	10.3	11.8	11.2	12.4	14.8	14.7	13.9	13.2	13.1
		MCWBR	5.6	6.0	5.7	5.8	7.1	7.5	8.4	8.8	7.5	6.5	5.9	6.0
(40) Clear-Sky Solar Irradiance	taub		0.403	0.396	0.384	0.361	0.342	0.331	0.316	0.324	0.353	0.382	0.393	0.397
	taud		2.517	2.540	2.557	2.606	2.614	2.637	2.661	2.625	2.531	2.476	2.482	2.503
	Ebn at Noon		299	297	293	287	281	279	287	295	298	298	300	301
	Edh at Noon		36	35	33	29	27	25	25	28	33	36	37	36
(44) All-Sky Solar Radiation	RadAvg		2011	1890	1744	1545	1331	1181	1336	1606	1947	2149	2193	2182
	RadStd		150	207	166	148	85	116	101	111	134	123	252	230

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page